

MOMENTUM AND REVERSALS

An In-Depth Analysis of DFA's Approach
Understanding the Short-Term vs. Long-Term Dynamic

ABSTRACT

This document provides a comprehensive examination of momentum effects in equity markets and Dimensional Fund Advisors' distinctive approach to managing this phenomenon. Unlike traditional momentum strategies that actively buy winners and sell losers, DFA uses momentum as a trading cost management tool, delaying purchases of securities with negative momentum and delaying sales of securities with positive momentum. This analysis explores the theoretical foundations, empirical evidence, implementation challenges, and philosophical reasons why DFA does not pursue momentum as a standalone premium despite its historical strength.

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1. MOMENTUM DEFINED: THE RELATIVE EFFECT

What is Momentum?

Momentum is fundamentally a **relative effect** that describes the tendency of securities to continue their recent performance patterns over short-term horizons:

- Securities that **outperform** other securities tend to continue to outperform in the short term (1-12 months)
- Securities that **underperform** other securities tend to continue to underperform in the short term (1-12 months)

Key Characteristics:

Characteristic	Description
Relativity	Momentum is measured relative to other securities, not in absolute terms. A stock with 5% returns might have negative momentum if peers returned 15%.
Time Horizon	Primarily a short-term effect, strongest over 1-12 month periods. Typical momentum strategies have 150-400% annual turnover.
Persistence	Recent winners tend to continue winning, recent losers continue losing, until the trend reverses (typically 3-8 months).
Cross-Sectional	Compares securities against each other, not against their own historical performance.

Academic Foundation:

The momentum effect was formally documented by Jegadeesh and Titman (1993) in their seminal paper *'Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency'* (Journal of Finance 48(1): 65-91). Their research demonstrated that strategies buying past winners and selling past losers generated significant abnormal returns over 3-12 month holding periods.

2. THE CHALLENGE: COMPETING PREMIUMS WITH DIFFERENT TIME SCALES

The central challenge in portfolio management is incorporating competing premiums that operate on fundamentally different time scales. This creates complex tradeoffs that require sophisticated implementation.

Time Scale Comparison

Premium	Time Scale	Typical Turnover	Average Holding Period
Momentum	1-12 months	150-400% annually	3-8 months
Value	3-5 years	~20% annually	~5 years
Size	3-5 years	~15-25% annually	4-6 years
Profitability	2-4 years	~25-35% annually	3-5 years

The Core Dilemma

When implementing value or small cap strategies, portfolio managers face a fundamental tension:

Long-Term Goal	Short-Term Reality	Resolution
Capture value premium over 3-5 year horizon	New value stocks often have negative momentum that persists 3-8 months	Delay purchases until momentum dissipates
Maintain exposure to value securities	Stocks leaving value often have positive momentum for 3-8 months	Delay sales to benefit from upward momentum
Low turnover, patient capital deployment	Momentum effects are stronger than value in short term	Use momentum screens to manage timing, not as trading signal

Critical Insight: The momentum effect is *stronger than the value effect in the short term*. This means that immediately buying a new value stock with downward momentum, or immediately selling a stock leaving value with upward momentum, could reduce or eliminate the expected benefits of value exposure during the transition period.

3. DOWNWARD MOMENTUM AND VALUE/SMALL CAP STOCKS

Why Value Stocks Often Have Negative Momentum

Understanding the relationship between value characteristics and momentum requires recognizing how stocks become value or small cap stocks in the first place:

The Value Creation Mechanism: Stocks generally become value stocks (high book-to-market, low price multiples) because they have *underperformed relative to other companies* in the market. This underperformance drives down their price relative to fundamentals like book value, earnings, or sales.

Similarly: Stocks become small cap stocks because their market capitalization has declined relative to the market, often due to underperformance.

The Momentum Implication

Step	Process	Momentum Status
1	Company experiences operational challenges, disappointing earnings, or negative news	Underperformance begins
2	Stock price falls relative to peers and fundamentals (book value, earnings)	Negative momentum develops
3	Price decline is large enough that stock now appears cheap on valuation metrics	Becomes eligible for value strategy
4	Value portfolio manager considers purchase	BUT: Still has negative momentum
5	If momentum persists, stock likely to continue underperforming for next 3-8 months	Expected near-term underperformance

DFA's Response: Delayed Purchases

To manage this competing premium dynamic, DFA employs a momentum screen that delays purchases of securities with negative momentum:

Without Momentum Screen	With DFA Momentum Screen
Action: Immediately buy stock when it becomes eligible for value strategy	Action: Delay purchase until negative momentum dissipates (typically 3-8 months)
Risk: Suffer momentum drag during transition period	Benefit: Avoid worst of continued underperformance
Result: Short-term losses may offset long-term value premium	Result: Better entry point, reduced volatility, improved risk-adjusted returns
Portfolio Impact: Higher short-term volatility and drawdowns	Portfolio Impact: Smoother return profile, lower turnover

Quantifying the Delay

According to Gerard O'Reilly's 2009 DFA paper 'Momentum Screens: A Three-Year Study,' delaying purchases by 3-8 months deals with momentum effects while adding value to small and value strategies. This delay period aligns with the typical persistence of momentum effects and the average holding period of securities in momentum strategies (150-400% turnover implies 3-8 month holding periods).

4. UPWARD MOMENTUM AND PORTFOLIO TURNOVER

The Exit Problem: When Stocks Leave Value

Just as there's a momentum challenge when stocks enter a value strategy, there's a symmetric challenge when stocks exit. The dynamics are reversed, creating an opportunity rather than a risk:

Stage	What Happens	Momentum Status	Traditional Response	DFA Response
Stock is in value portfolio	Trading at low valuation multiples	May have negative or neutral momentum	Hold position	Hold position
Company improves: better earnings, positive news	Stock price rises relative to peers	Begins developing positive momentum	Consider selling if no longer "value"	Continue monitoring
Price rises enough that stock no longer meets value criteria	Stock now has higher valuation multiples	Has positive momentum	Sell immediately to buy other value stocks	Delay sale to capture momentum premium
3-8 months later	Momentum effect has dissipated	Momentum neutral	N/A - already sold	Now sell, buy other value stocks

The Opportunity Cost of Immediate Selling

When a stock leaves the value universe, it typically has **upward momentum** because it outperformed its peers. The opportunity cost of immediately selling such a stock is the *upward momentum premium* that could be captured over the next 3-8 months.

Because the short-term effect of momentum is *larger than the value effect*, the portfolio is better off:

1. **Delaying the sale** of stocks with positive momentum that are leaving value
2. **Benefiting from** the expected continued outperformance (momentum premium)
3. Rather than **immediately reinvesting** in new value stocks that may have negative momentum

Key Insight: Time Scale Dominance

This approach works because of a critical empirical observation:

Time Horizon	Dominant Effect	Implication
1-12 months (Short term)	Momentum > Value Momentum effects are stronger	Use momentum screens to improve timing of value trades
3-5 years (Long term)	Value > Momentum Value effects dominate, momentum mean-reverts	Structure portfolio around long-term value exposure, use momentum for tactical timing

5. DFA'S DISTINCTIVE APPROACH: MOMENTUM AS A TRADING COST

Not a Momentum Strategy

It is crucial to understand what DFA is **not** doing:

- **NOT** following a traditional momentum strategy that buys stocks with upward momentum and sells stocks with downward momentum
- **NOT** actively trading to capture momentum premiums as a standalone dimension
- **NOT** using momentum as a primary portfolio structuring principle

What DFA IS Doing

Instead, DFA incorporates momentum effects into existing strategies by:

Action	Purpose	Impact
Delay purchases of securities with negative momentum	Avoid providing liquidity to momentum traders who are selling (and driving prices down)	• Better entry prices • Avoid momentum drag • Lower short-term volatility
Delay sales of securities with positive momentum	Avoid providing liquidity to momentum traders who are buying (and driving prices up)	• Better exit prices • Capture upward momentum • Reduced opportunity cost
Maintain consistency with value/small/prof strategies	Preserve overall portfolio characteristics and long-term premium exposures	• No increase in turnover • Maintains diversification • Preserves factor tilts

The 'Avoiding Providing Liquidity' Concept

This is perhaps the most important conceptual insight into DFA's approach. When momentum traders are actively selling a stock (creating downward momentum), someone must buy from them. When they're actively buying (creating upward momentum), someone must sell to them. By delaying its trades, DFA:

1. **Avoids being the counterparty** that provides liquidity to momentum traders at unfavorable prices
2. **Lets momentum effects play out** before entering or exiting positions
3. **Trades after price pressure subsides**, getting better execution

Benefit Without Cost

The elegant aspect of this approach is that it allows DFA to:

- **Enjoy benefits** of upward momentum without increasing turnover
- **Insulate portfolios** from downward momentum without changing strategy
- **Add value** over a shorter time scale (3-8 months) than value itself (3-5 years)

- **Sacrifice nothing** in terms of overall characteristics or factor exposures

6. THE MAGNITUDE OF THE MOMENTUM PREMIUM

Historical Evidence of the Momentum Effect

Despite DFA's decision not to pursue momentum as a standalone premium, the historical magnitude of the momentum effect is substantial and well-documented:

Period	Market Segment	Momentum Premium	Benchmark
1927-2015	Large Caps	~4.0% annually	Long side minus S&P 500
1927-2015	Small Caps	~6.0% annually	Long side minus CRSP 6-10
1927-2016	Overall Market	~8.1% annually	Winner minus Loser portfolio

Source: Kenneth R. French Data Library (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html)

Why Such Large Historical Returns?

The magnitude of historical momentum premiums naturally raises the question: Why doesn't DFA pursue this more aggressively? Several factors explain both the size of historical premiums and the caution around future expectations:

Factor	Explanation
Pervasiveness	Momentum appears across virtually all asset classes, geographies, and time periods, suggesting a robust phenomenon
Persistence	The effect has persisted even after academic publication and widespread knowledge, unlike many other anomalies that disappear after discovery
Behavioral Foundation	May be driven by slow information diffusion, underreaction to news, or herding behavior - but these explanations remain debated
Implementation Costs	High turnover (150-400% annually) creates substantial transaction costs that eat into gross returns
Risk of Crashes	Momentum strategies are vulnerable to sudden, severe reversals during market turmoil (e.g., 2009 saw momentum crash of -50%+)

7. WHY MOMENTUM FAILS THE SENSIBLENESS CRITERION

DFA's Three Criteria for Factor Premiums

Dimensional employs three criteria when evaluating whether to structure investment solutions around a factor premium:

Criterion	Definition	Momentum Status
Persistent	Premium has existed across long time periods	✓ PASS Present 1927-2016 (90+ years)
Pervasive	Premium appears across markets, geographies, asset classes	✓ PASS Global, across equities, bonds, commodities, currencies
Sensible	We understand WHY it exists, what risk it compensates for, or what economic mechanism drives it	✗ FAIL No consensus on why momentum exists

The Sensibleness Problem

While momentum clearly exists in the data, we lack a compelling explanation for *why* it exists. This creates several concerns:

Concern	Implication
No Risk Story Unlike value (distress risk) or size (liquidity/information risk), momentum doesn't clearly compensate for a specific risk exposure	If not risk-based, why does the market allow it to persist? This raises questions about whether it will continue.
Behavioral Explanations Weak Explanations like "underreaction" or "herding" are vague and don't specify precise mechanisms	Behavioral biases can be arbitrated away once known. Why hasn't momentum been arbitrated away?
Technology/Liquidity Changes Some academics suggest momentum persists because trading costs were too high to exploit it profitably	As markets become more liquid and trading costs fall, will momentum disappear? This threatens future premiums.
Crash Risk Momentum strategies experience sudden, severe crashes that are difficult to explain or predict	Without understanding the mechanism, we can't predict or manage these crashes. This makes risk management challenging.

Quote from the Document:

"Without a sensible theoretical underpinning, we must carefully consider the risk that momentum premiums will not persist in the future. Some academics have suggested that momentum survives because it is too costly for managers to actively trade on momentum. If this is the case, will improvements in market liquidity and trading technologies impact

future momentum premiums?"

8. IMPLEMENTATION CHALLENGES AND COST CONSIDERATIONS

The High Cost of Momentum Strategies

Even if momentum premiums persist, capturing them through active momentum strategies faces substantial implementation challenges:

Cost Category	Magnitude	Impact on Net Returns
Turnover	150-400% annually	Each 100% turnover might cost 0.5-1.0% in transaction costs
Bid-Ask Spreads	Higher for small caps, emerging markets	Spreads widen during volatile periods when momentum strategies trade most
Market Impact	Significant for large implementations	Large momentum trades move prices adversely, reducing returns
Short-Selling Costs	Required for traditional long-short momentum	Borrow costs, short squeeze risk, dividend payments on short positions
Taxes	High turnover creates short-term capital gains	For taxable investors, after-tax returns could be significantly impaired

Empirical Question: Can Momentum Be Captured Cost-Effectively?

The document notes: *"It remains an open empirical question whether momentum can be captured cost-effectively in broadly diversified strategies."*

This is a critical point. The historical gross premium of ~8% annually must overcome:

- Transaction costs: ~1.5-4.0% annually (at 150-400% turnover)
- Management fees: ~0.5-1.0% annually
- Slippage and market impact: ~0.5-2.0% annually
- Potential taxes: Could eliminate remaining returns for taxable investors

Net Result: Even a strong 8% gross premium could become 1-3% net, or even negative after all costs. If the premium fails to materialize in the future (due to lack of sensible foundation), the high ongoing costs become an especially heavy burden.

Comparison: DFA's Low-Cost Momentum Integration

Approach	Turnover Impact	Cost to Capture	Downside if Disappears
Traditional Momentum Strategy	Creates 150-400% annual turnover	High: 2-5% annually in all-in costs	Severe: Costs persist, no offsetting returns
DFA Momentum Screens	Doesn't increase turnover	Very Low: Only opportunity cost of delays	Minimal: No extra costs incurred, portfolios still well-designed

9. THE TIME SCALE PROBLEM

Fundamental Incompatibility of Time Horizons

One of the deepest challenges in combining momentum with value/size/profitability strategies is the fundamental incompatibility of their time scales:

Dimension	Momentum	Value/Size/Profitability
Optimal Rebalance	Monthly or even weekly to capture short trends	Annually or less frequently to reduce costs and taxes
Average Holding Period	3-8 months before trend reverses	3-5 years for premiums to materialize
Portfolio Turnover	150-400% annually	15-30% annually
Primary Risk	Sudden trend reversals (momentum crashes)	Long periods of underperformance (value traps)
Return Predictability	Strong over 1-12 months	Weak over 1-12 months, strong over 3-5 years

The Integration Challenge

Combining factors with such different time scales creates several problems:

- 1. Conflicting Signals:** Momentum might say "sell" when value says "strong buy." Which do you follow?
- 2. Turnover Explosion:** Pursuing both simultaneously dramatically increases turnover and costs
- 3. Risk Management Complexity:** Managing drawdown risk becomes extremely complex with mixed time horizons
- 4. Client Communication:** Explaining short-term momentum-driven trades in a long-term value strategy is difficult

DFA's Solution: Use Momentum on Value's Time Scale

Rather than trying to capture momentum as a standalone premium, DFA uses momentum information to *improve the timing* of value/size trades:

Problem	DFA Solution
Momentum requires monthly rebalancing	Use momentum screens only when already trading for value/size reasons
Momentum holding period is 3-8 months	Delay trades by 3-8 months, aligning with momentum dissipation timeline
Momentum increases turnover	Screens delay rather than induce trading, so turnover stays low

Difficult to explain mixed strategies	Primary strategy remains value/size/prof; momentum is just implementation detail
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This approach allows DFA to "add value over a shorter-term time scale than the value time scale" without fundamentally changing the portfolio's structure or characteristics.

10. DIMINISHING MARGINAL BENEFITS

The Portfolio Construction Context

A critical but often overlooked consideration: DFA doesn't evaluate momentum in isolation. They evaluate it in the context of portfolios that *already* pursue multiple premiums:

Factor Already Pursued	Annual Premium (Historical)	Implementation Cost	Current Portfolio Exposure
Size (Small Cap)	~3-4%	Low (~20% turnover)	High tilt in DFA small cap strategies
Value (HML)	~4-5%	Low (~20% turnover)	High tilt in DFA value strategies
Profitability (RMW)	~3-4%	Medium (~30% turnover)	Integrated into core strategies
Momentum Screens (Current Approach)	~1-2% (estimated)	Very Low (no added turnover)	Already implemented via delays

The Marginal Benefit Question

Given this context, the question becomes: What is the *marginal* benefit of pursuing momentum more aggressively?

Consideration	Analysis
Already Capturing Some Momentum Benefits	Via existing screens, DFA estimates they capture 1-2% of the ~8% gross momentum premium with negligible cost
Correlation with Existing Factors	Momentum has negative correlation with value (-0.3 to -0.5). Adding more momentum reduces value tilt, potentially undoing existing exposures.
Marginal Cost	Moving from screens to active momentum would require 130-380% additional turnover, adding 1.5-3.5% in costs
Marginal Premium	Incremental gain would be ~6% gross (8% total - 2% captured), but after 2-4% in costs, net marginal gain might be only 2-4%
Risk Trade-off	Would introduce momentum crash risk without clear understanding of when/why crashes occur

Quote from the Document:

"Because we already pursue size, value, and profitability premiums in our strategies and consider momentum in how we manage turnover, the marginal contribution after costs of the momentum premium would likely be very small or even negative, especially considering the high turnover required to pursue that premium and the fact that there is no expectation of a momentum premium in the first place."

This is perhaps the most sophisticated argument against aggressive momentum pursuit: it's not that momentum doesn't exist or isn't valuable in isolation, but that its *marginal* contribution to an already well-diversified,

multi-factor portfolio is questionable after considering all costs and risks.

11. EMPIRICAL EVIDENCE AND ACADEMIC RESEARCH

Key Research Supporting Momentum

The academic literature on momentum is extensive. Key papers include:

Authors	Paper	Key Finding
Jegadeesh & Titman (1993)	Returns to Buying Winners and Selling Losers	First documentation of momentum profits: 3-12 month strategies earned ~1% monthly
Fama & French (2015)	Incremental Variables and the Investment Opportunity Set	Momentum adds explanatory power beyond size, value, profitability, and investment
O'Reilly (2009) DFA Paper	Momentum Screens: A Three-Year Study	DFA's screens add value to small/value strategies without increasing turnover
Asness, Moskowitz & Pedersen (2013)	Value and Momentum Everywhere	Momentum works across asset classes, geographies, and time periods

Momentum Premium by Market Segment (US, 1927-2015)

Market Segment	Momentum Premium	Measurement Method	Correlation with Value
Large Caps	~4% annually	Long side minus S&P 500	-0.3 to -0.4
Small Caps	~6% annually	Long side minus CRSP 6-10	-0.4 to -0.5
Overall (Winner-Loser)	~8.1% annually	Top decile minus bottom decile based on past 2-12 month returns	-0.3 to -0.5

Source: Kenneth R. French Data Library. Negative correlation with value means momentum and value often move in opposite directions.

Momentum Crashes: The Dark Side

One of the most concerning aspects of momentum strategies is their vulnerability to sudden, severe crashes. Examples include:

Period	Market Context	Momentum Strategy Loss
2009 (Q2)	Post-financial crisis recovery; sharp reversal from extreme pessimism	-50% to -70% (worst losers became winners)
1933	Great Depression recovery; extreme reversal from distressed markets	-40% to -60%
2001 (Post-9/11)	Market recovery from tech bubble burst and 9/11 panic	-25% to -35%
Various bear-to-bull transitions	Momentum crashes occur when markets rapidly reverse from extreme conditions	Typically -20% to -50% over 1-3 months

These crashes are particularly problematic because: (1) they're unpredictable without understanding momentum's mechanism, (2) they can wipe out years of accumulated returns in weeks, and (3) they often occur precisely when investors are most vulnerable (market stress periods).

12. CONCLUSION: INTEGRATION VS. ISOLATION

Summary of DFA's Approach

Dimensional's treatment of momentum reflects a sophisticated understanding of factor premiums, implementation costs, and portfolio construction principles. Rather than viewing momentum as a standalone dimension to pursue aggressively, DFA integrates momentum considerations into their existing value/size/profitability strategies.

Aspect	DFA Position	Rationale
Momentum as Asset Allocation	Do NOT use momentum for asset allocation decisions	Lacks sensible foundation; unclear if premiums will persist
Momentum in Implementation	DO use momentum screens to manage trading timing	Costs are minimal; benefits persist even if momentum disappears
Primary Strategy	Structure portfolios around value, size, profitability	These have sensible risk-based or economic foundations
Momentum's Role	Tactical tool to improve trade execution	Delays protect against momentum drag; don't change core exposures
Turnover Impact	Momentum screens REDUCE effective turnover costs	Better timing means better prices without additional trades
Risk Management	Avoid momentum crash risk	No active momentum exposure means no vulnerability to crashes

The Philosophical Core

At its heart, DFA's momentum philosophy rests on a simple but profound insight:

"We don't use momentum as part of our asset allocation decisions, but we don't ignore it in our investment process, either."

This middle path acknowledges momentum's empirical reality while respecting the limits of our understanding. It captures benefits where they're available at low cost, while avoiding the high-cost, high-risk proposition of pursuing momentum as a primary strategy.

Final Comparative Framework

Question	Traditional Momentum Strategy	DFA Momentum Screens
Do we believe momentum exists empirically?	YES - 8% historical premium	YES - 8% historical premium
Do we understand WHY it exists?	Maybe behavioral, but uncertain	NO - fails sensibleness test
Can we capture it cost-effectively?	Uncertain; high costs (2-5% annually)	YES - very low costs (<0.5% annually)

What if it disappears in the future?	Major problem: high costs persist, no returns	Minor problem: negligible costs, still have value/size
Impact on portfolio turnover?	Massive increase (+150-400%)	No increase (or slight decrease)
Risk of momentum crashes?	Direct exposure; vulnerable	No direct exposure; protected
Alignment with long-term factors?	Conflicts with value (negative correlation)	Enhances value by improving timing
Client communication complexity?	High - mixed messages about time horizon	Low - still primarily a value/size story

CONCLUDING THOUGHTS

DFA's nuanced approach to momentum exemplifies their broader investment philosophy: use market premiums where they're available at reasonable cost with sensible foundations, but don't chase premiums simply because they appear in historical data.

The momentum premium is real, substantial, and persistent. But its lack of a sensible theoretical foundation, combined with high implementation costs and crash risk, makes aggressive pursuit questionable. By incorporating momentum through low-cost screens rather than high-turnover strategies, DFA achieves an elegant solution: capture some momentum benefits while maintaining focus on the dimensions (value, size, profitability) that have both empirical support AND sensible explanations.

This approach demonstrates the importance of not just identifying premiums in data, but understanding their sources, costs, and interactions with other portfolio components. It's a masterclass in the difference between data mining and principled investing.

Key Lessons for Exam Preparation

1. **Momentum is real but not sensible:** It passes persistence and pervasiveness tests but fails sensibleness
2. **Time scales matter:** Momentum (1-12 months) operates on a different scale than value (3-5 years)
3. **DFA uses momentum defensively:** Screens delay trades to avoid momentum drag, not to actively capture momentum
4. **High turnover = high costs:** 150-400% turnover makes net momentum capture questionable
5. **Marginal vs. absolute benefits:** Adding momentum to existing value/size/prof strategies has low marginal benefit
6. **Risk-adjusted thinking:** Momentum crashes can wipe out years of gains without warning
7. **Integration over isolation:** Consider how premiums interact, not just their standalone magnitude

Prepared for DFA Milestone 1 exam. Source: Principles of Dimensional Investing, Equity Strategies section on Momentum (pages 49-51) and More on Momentum sidebar (page 51).